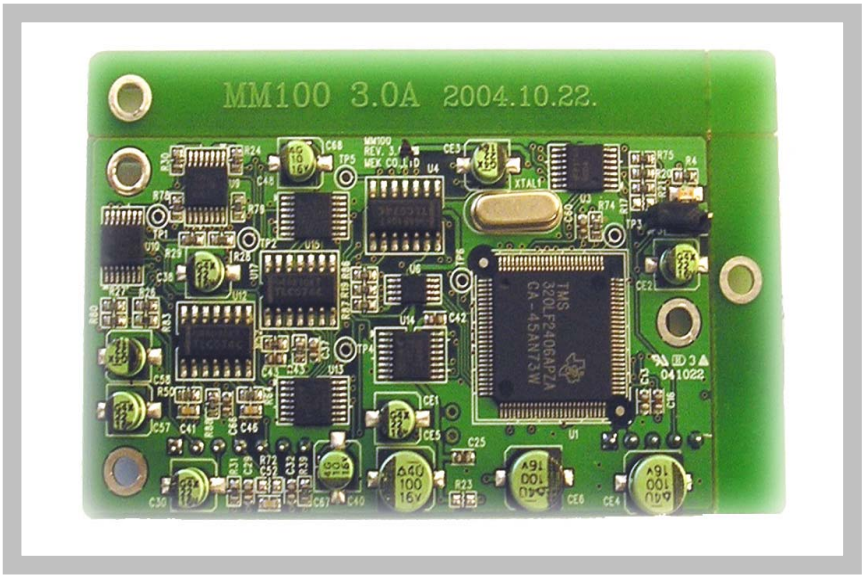
MEKICS *Intensive Care System*

User's Manual

MM100plus



SpO_2 Module

MEK
Intensive Care System

1. Specification

Sensor	*Finger Type Reusable Sensor*	
	(MEK Type / Nellcor Type both Available)	
SpO2 Percentage Oxygen Saturation	Range of Display	*0~100%*
	Resolution	*1%*
	Accuracy	*±2%, For Reading Between 70~100%* *±3%, For Reading Between 50~69%*
	Setting Time	*Wave Data Out Time : 2sec max* *Percentage Data Out Time : 5sec max*
	Average	*4 Time Average*
Pulse Rate	Range of Display	*30~250 BPM*
	Resolution	*1 BPM*
	Accuracy	*±2%*
	Averaging	*4 Times Average*
Output	*RS-232C Baud Rate 19200bps, no partity, 8 bit, 1 stop bit standard*	
	SpO2, Pulse rate, Infra-red wave data, Status Message	
Power	*DC + 8.0 ~ +10.0 Ripple & Noise ±5% / 125mA Type*	
Power Consumption	*1W Type, 125mA / +8.0V type in Probe operation*	
Dimensions	*W75 × D40 × H15 (mm)*	
Features	*Single Power +8.0V*	
	Direct PC Interface of RS232	
	Fast Response Time under 5 second for patient moving	

2. Description of Model Name (MM100p-xx)

2-1. High ripple noise source type

Model name	Description
MM100P-S	std RS232 (+/-12V signal direct PC interface), +/-12V Level
MM100P-T	TTL level com port (H: +3.3V, L: 0V), +/-3.3V Level

S : IC in backside of MM100p

T : No IC in backside of MM100p

3. Communication protocol

2-1. H/W protocol

RS-232C, 19200 bps, 8 bits, 1 stop bit, No parity, CTS/RTS not used

2-2. S/W protocol structure

Data structure, there is 2-types.

1) Standard type

MSB ------- LSB

First byte : 1 h h h h d d d

Second byte : 0 d d d d d d d

Third byte : 1 1 1 1 1 1 X X X (0xF8 ~ 0xFF) : Tail code

hhhh : 4 Bit Header information

0010	SpO2 Wave data
0110	SpO2 pulse rate data
0111	SpO2 saturation data
1101	2 Byte expended type

dddddddddd : 10 Bit Data

2) Extended header type

MSB ------- LSB

First byte : 1 1 1 0 1 h h h

Second byte : 0 d d d d d d d

Third byte : 1 1 1 1 1 1 X X X (0xF8 ~ 0xFF) : Tail code

hhh : extended header

001	Probe ID 0 : MEK Unique type 1 : Nellcor compatible type
100	Pulse wave Gain information (0 ~ 7)
110	System Information 0x01: Power On 0x02: Power Off 0x09: Event PR (PR detected event) 0x11: Standby mode 0x12: Search mode 0x13: check probe error (No Probe, week receive gain) 0x14: Normal

ddddddd : Data bit

2-3. Protocol Table: MM100 <-> Host

Data type S: standard form, E: extended header type

T: MM100 → host, R: host → mm100

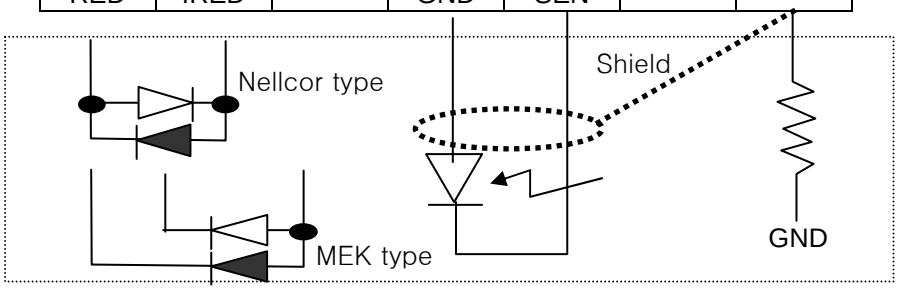
	First Byte	Second Byte	Data Occurring	Data
	MSB <-------> LSB	MSB <-------> LSB	Rate/Index	Type
SpO2 Wave	1 0 0 1 0 d d d	0 d d d d d d d	120Hz	S,T
SpO2 PR	1 0 1 1 0 d d d	0 d d d d d d d	30bpm ~ 250bpm	S,T
SpO2 Value	1 0 1 1 1 d d d	0 d d d d d d d	30bpm ~ 250bpm	S,T
System Information	1 1 1 0 1 1 1 0	0 0 0 0 0 0 0 1	0x01: Power On	E,T
		0 0 0 0 1 0 0 1	0x09 : Event PR	E.T
		0 0 0 1 0 0 0 1	0x11:Stb mode	E,T
		0 0 0 1 0 0 1 0	0x12:Search mode	E,T
		0 0 0 1 0 0 1 1	0x13: check probe Error (No Probe or Week receive gain)	E,T
		0 0 0 1 0 1 0 0	0x14: Normal mode	E,T

** System information is only the status of module. But this information is useful for user interface.

4. Interface Connector

4-1. PROBE INTERFACE CONNECTOR

1	2	3	4	5	6	7
TX_ RED	TX_ IRED	VCC	RX_ GND	RX_ SEN	P_ID	GND



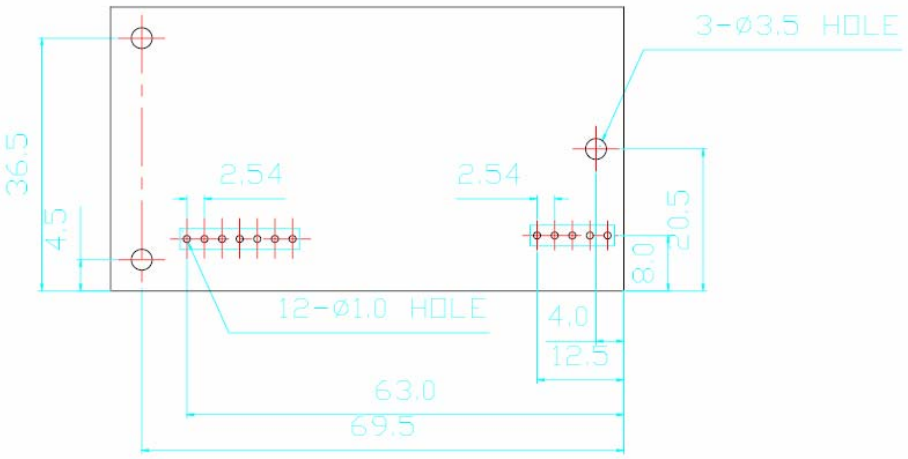
** P_ID: MEK or Nellcor compatible

4-2. POWER & COMMUNICATION CONNECTOR

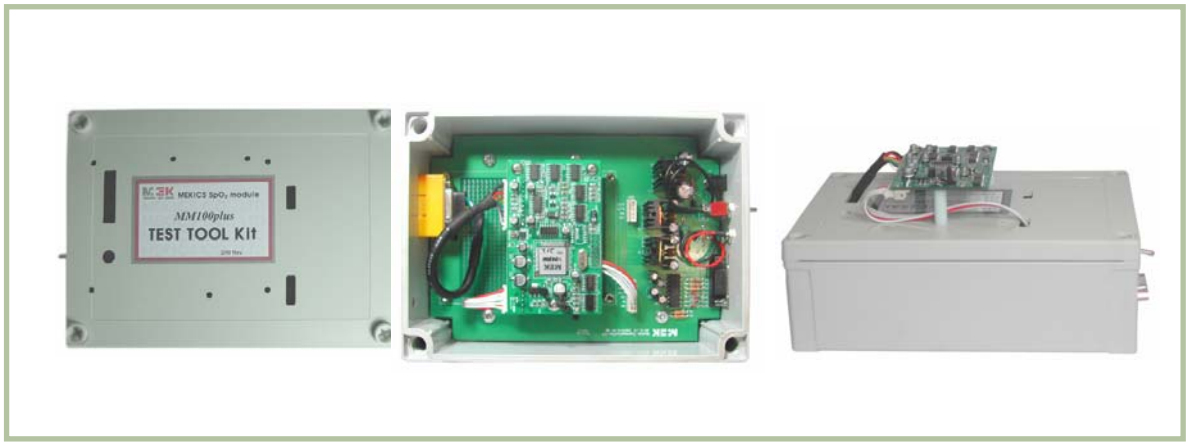
1	2	3	4	5
TX	RX	+VCC (7.5V)	NC	GND

4. Dimension

4-1 : (W)82.5 x (H)50 x (D)17mm



5. List of MM100plus Test Took Kit (Optional)

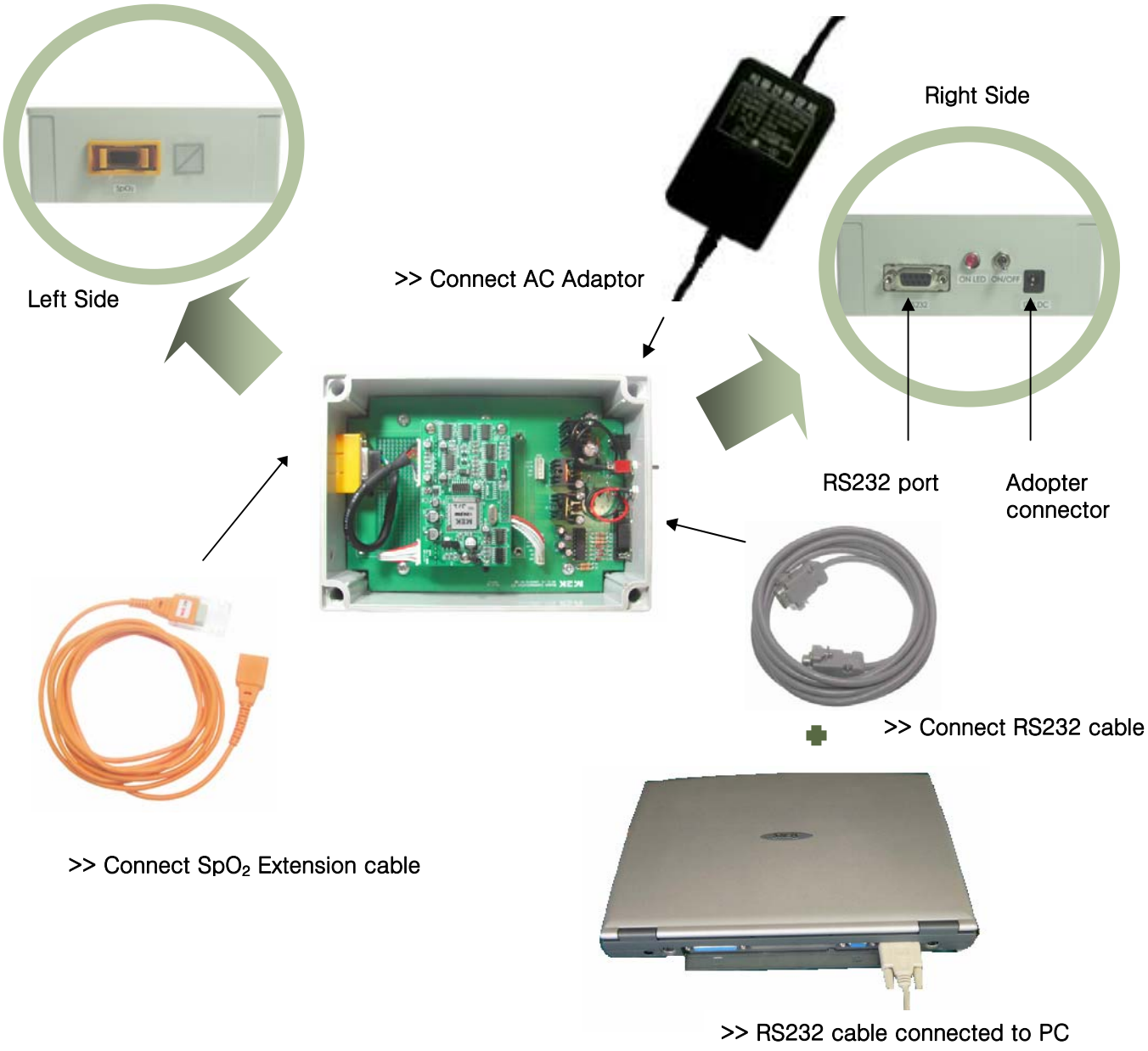


NO	Part	Description / Shape
1	MM100 Plus module (Installed)	 >> Front Side  >> Back Side
2	Communication Jig (Installed)	 → AC Adapter → Power → RS232 port

NO	Part	Description / Shape
3	SpO2 Sensor (Adult)	
4	Sensor Extension Cable (2.5M)	
5	RS-232 Cable (Connect PC & Jig)	
6	AC Adapter (220V AC) ** 12V / 1A / 50Hz	
7	PC test Software	3.5″ diskette or Provided by On-line
8	Technical manual	Printed or Provided by On-line

** The Length of Sensor Extension Cable cannot be exceeding the 2.5M.

6. Test Kit Assembling

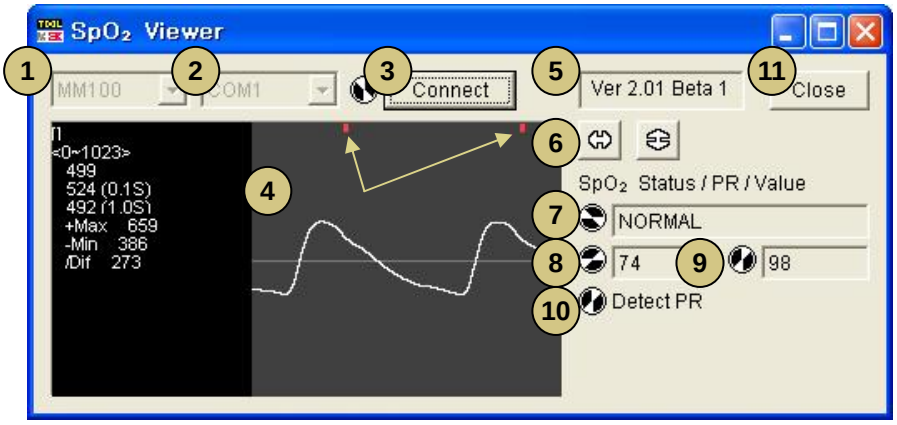


>> When you finish the assembling, please connect SPO2 sensor (part No.3) to Extension cable (part No. 4)

>> Plug in Adaptor and then make power on in Test Tool kit

7. How to Use Test Program 'ViewerSPO'

1) Complete the Test Tool Kit Assembling
2) Turn on the power of Test Tool Kit (Power Key in right side)
(Make sure the light is on in LED when turn on the power)
3) Double click "ViewerSPO.exe" test program in your PC
4) Select COM port(COM1 or COM2) which you want, then click " Connect"
5) If the program is working collect, you should see the straight line in screen
If not, please check the Test Kit Assembling again and reset-up.
6) Connect SpO2 sensor to patient's Finger and you can see message of module status, Pulse Rate, SpO2 Value in program view.



>> ViewerSPO Test Program

No.	Description
①	Type of Module (MM100)
②	Select Com port (COM1 or COM2)
③	Start Testing
④	Sp02 wave and Red marking (Red marking is detection point of PR)
⑤	Version of Sp02 module
⑥	Control the size of viewer
⑦	Status of Moudle
⑧	Pulse Rate
⑨	Sp02 Value (percentage)
⑩	Status of detection of PR(Pulse Rate)
⑪	Close Testing

Contact Us

▶ **Module Sales Department.**

e-mail: simo@mek-ics.com

Tel: 82-33-264-5926 (Ext. 641)

Fax: 82-33-264-5999

Name: Sang-il Mo /Sales Manager